

GEE Lab - 1

WORKING WITH GOOGLE EARTH ENGINE

TO SEARCH, FIND AND VISUALIZE A BROAD RANGE OF REMOTELY
SENSED DATASETS

Free cloud processing with built-in functions



Colocated Data + Computation + APIs

GOOGLE EARTH ENGINE

A CLOUD-BASED PLATFORM FOR PLANETARY-SCALE GEOSPATIAL ANALYSIS

No more downloading and analyzing individual tiles at a time

Sign up for a Google Earth Engine account

Earth Engine account registration: <https://signup.earthengine.google.com/#!/>

Once registered you can access the Earth Engine environment @:
<https://code.earthengine.google.com>

Google Earth Engine Code Editor uses the JavaScript programming languages.

For JavaScript Background: https://developers.google.com/earth-engine/tutorial_js_01

Google Mission Statement

"To organize the world's information and make it universally accessible and useful."

Earth Engine provides an API to process, analyze, and visualize the data using Google machines.

Data storage and processing are all performed at Google.

Whatever USGS has, Earth Engine has.

All you need to use Earth Engine is an internet connection.

Earth Engine Data Catalog

- No more downloading and analyzing individual tiles at a time.
- No more worry about your local storage.

As of 2020

> 200 public datasets

> 5 million images

> 4000 new images every day

~ 17 petabytes of data

Now

- 1000 of public datasets
- Thousands of new images added every day
- Total Stored Data: Over 90 petabytes (PB) and expanding rapidly.
- Growth Rate: Grows by roughly 1 petabyte of new data every month.
- Historical Span: Over 50 years of continuous global satellite records.

The Earth Engine Data Catalog



Landsat & Sentinel 1, 2
10-30m, weekly



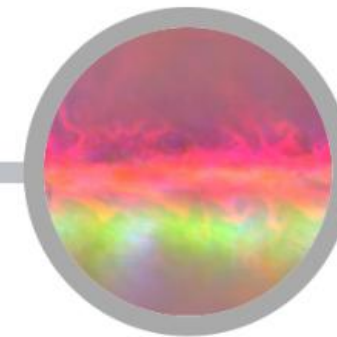
MODIS
250m daily



Vector Data
WDPA, Tiger



**Terrain &
Land Cover**



Weather & Climate
NOAA NCEP, OMI, ...

... and upload your own vectors and rasters

There is a one or two-day latency between scene acquisition and when it is ingested into the catalog.

33 years

Of satellite data

Over 5,000,000

Landsat and Sentinel scenes analyzed

3 Quadrillion Pixels

3,000,000,000,000,000

- ✓ Originally launched more than a decade ago for research; now available for commercial use

- ✓ Ability to integrate with powerful analytics and tools like BigQuery and Vertex AI
- ✓ Remains at no cost for nonprofit, noncommercial, and research projects; [learn more](#)

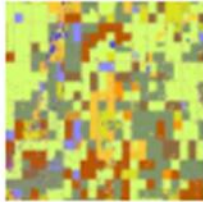
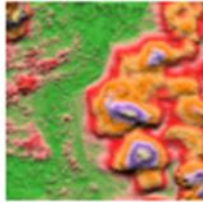
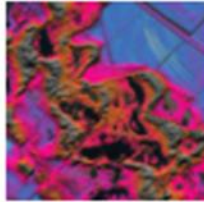
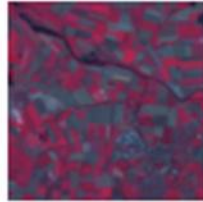
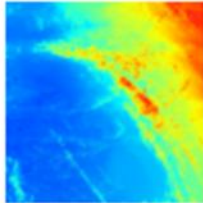
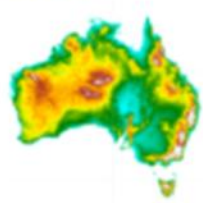
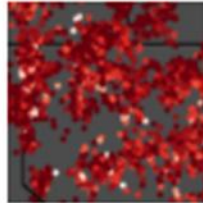
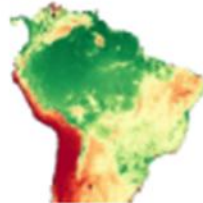
A planetary-scale platform for Earth science data & analysis

Earth Engine's public data archive includes more than forty years of historical imagery and scientific datasets, updated and expanded daily.

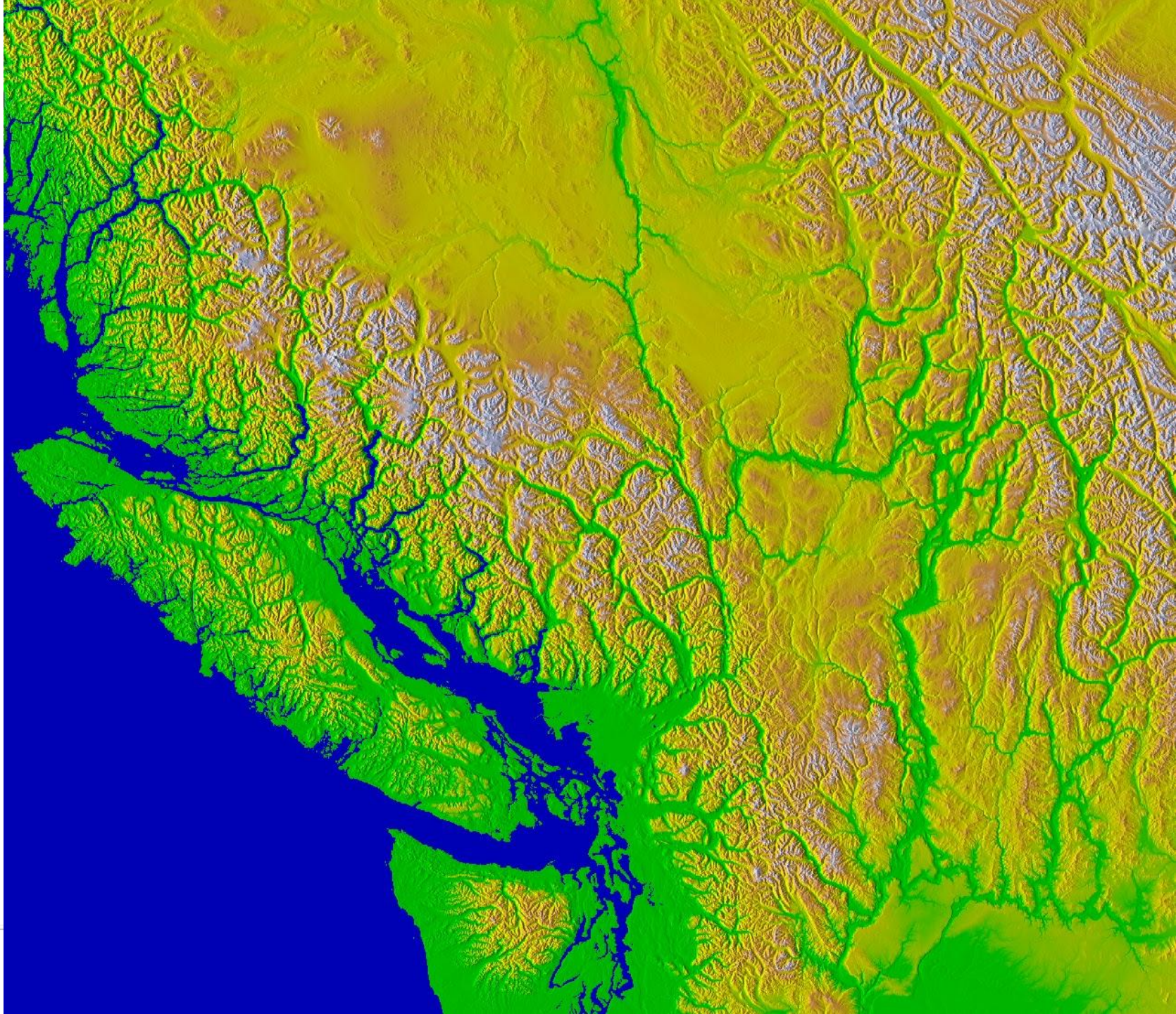
[View all datasets](#)

Earth Engine's public data catalog includes a variety of standard Earth science raster datasets. You can import these datasets into your script environment with a single click. You can also upload your own raster data or vector data for private use or sharing in your scripts.

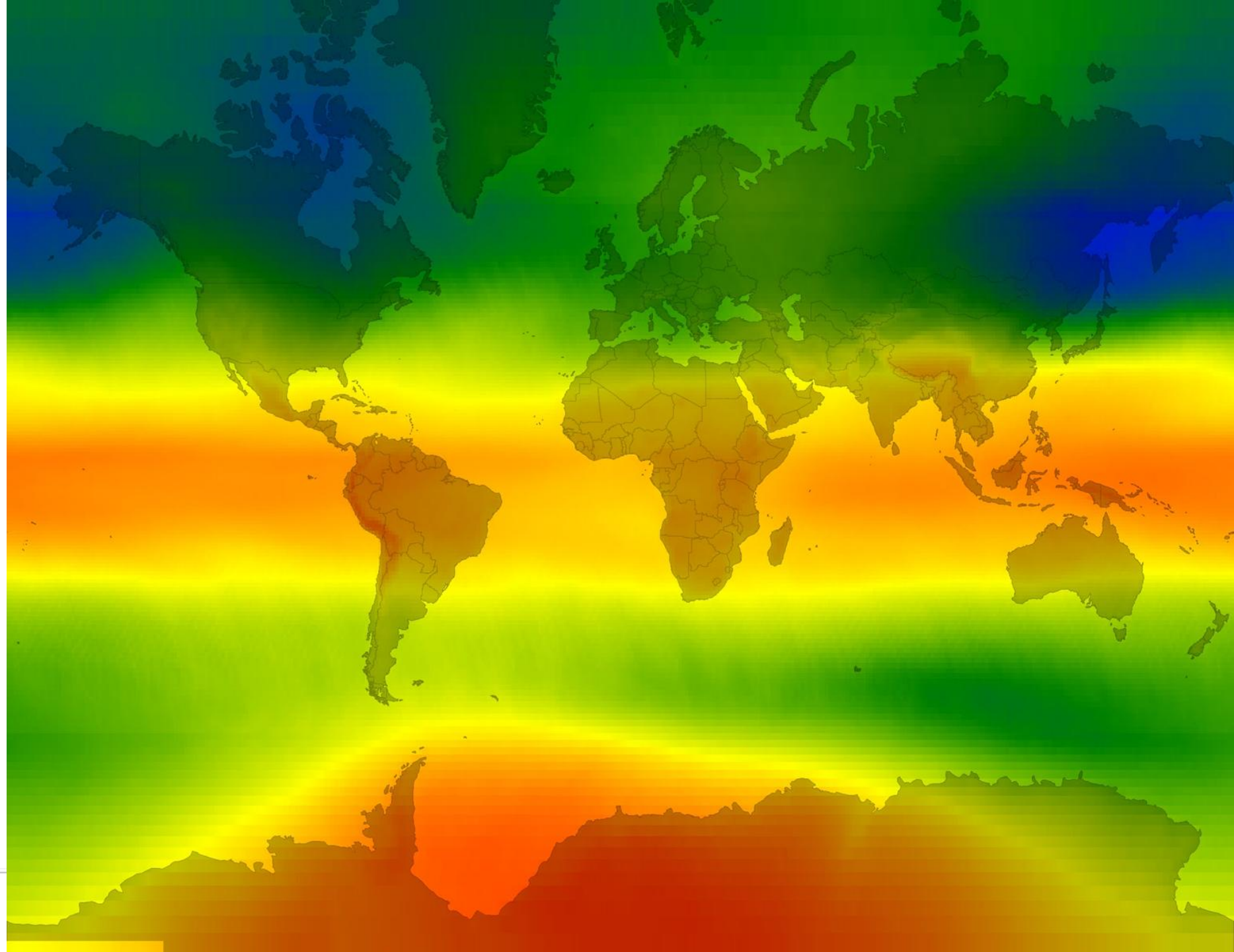
Looking for another dataset not in Earth Engine yet? Let us know by [suggesting a dataset](#).

Canada AHC Annual Crop Inventory  Starting in 2009, the Earth Observation Team of the Science and Technology Branch (STB) at Agriculture and Agri-Food Canada (AAFC) began the process of generating annual crop type digital maps. Following the Remote Sensing in 2009 and 2010's Decision Tree (DT) based methodology... img landsat canada see	AHN Netherlands 0.5m DEM, Interpolated  The AHN DEM is a 0.5m DEM covering the Netherlands. It was generated from LiDAR data taken in the spring between 2007 and 2012. It contains ground level samples with all other items above ground (such as buildings, bridges, trees etc.) removed. This version is... dem elevation netherlands dem geographical ahl	AHN Netherlands 0.5m DEM, Non-Interpolated  The AHN DEM is a 0.5m DEM covering the Netherlands. It was generated from LiDAR data taken in the spring between 2007 and 2012. It contains ground level samples with all other items above ground (such as buildings, bridges, trees etc.) removed. This version is... dem elevation netherlands dem geographical ahl	AHN Netherlands 0.5m DEM, Raw Samples  The AHN DEM is a 0.5m DEM covering the Netherlands. It was generated from LiDAR data taken in the spring between 2007 and 2012. This version contains both ground level samples and items above ground level (such as buildings, bridges, trees etc.). The point cloud... dem elevation netherlands dem geographical ahl	ASTER L1T Radiance  The Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) is a multispectral imager that was launched on board NASA's Terra spacecraft in December, 1999. ASTER can collect data in 14 spectral bands from the visible to the thermal infrared. Each scene covers an area of... radiance thermal see
Australian 5M DEM  The Digital Elevation Model (DEM) 5 Meters of Australia derived from LiDAR data represents a National 5 meter (bare earth) DEM which has been derived from some 225 individual LiDAR surveys between 2001 and 2012 covering an area in excess of 500,000 square kilometers... elevation australia dem geographical geosciencesaustralia ge	DEM-H: Australian 5M DEM Hydrologically Enforced Digital Elevation Model  The Hydrologically Enforced Digital Elevation Model (DEM-H) was derived from the SRTM data acquired by NASA in February 2000. The model has been hydrologically conditioned and drainage enforced. The DEM-H captures flow paths based on SRTM elevations and mapped stream lines, and supports delineation of... elevation australia dem geographical smoothed geosciencesaustralia	DEM-S: Australian Smoothed Digital Elevation Model  The Smoothed Digital Elevation Model (DEM-S) was derived from the SRTM data acquired by NASA in February 2000. DEM-S represents ground surface topography (excluding vegetation features) and has been smoothed to reduce noise and improve the representation of surface shape. An adaptive process applied more... elevation australia dem geographical smoothed geosciencesaustralia	BLM MAM TerraNet Semidigital AM Point v1  Since 2011, the Bureau of Land Management (BLM) has collected field information to inform land health through its Assessment Inventory and Monitoring (AIM) strategy. To date, more than 5,000 semidigital AIM field plots have been collected over BLM lands. The BLM AIM data archive is... aim landsat semidigitalam state assessment environment	Penman-Monteith-Lauring Bioclimatic Vegetation V2 (PMV2) products  Penman-Monteith-Lauring Bioclimatic Vegetation V2 (PMV2) products include evapotranspiration (ET), net stress component, and gross primary product (GPP) at 300m and 6-day resolution during 2000-2017 and with spatial range from 40°S to 90°N. The major advantages of the PMV2 products are: coupled estimates of transpiration and GPP... evapotranspiration grossprimaryproduct gpp gpi see gpm

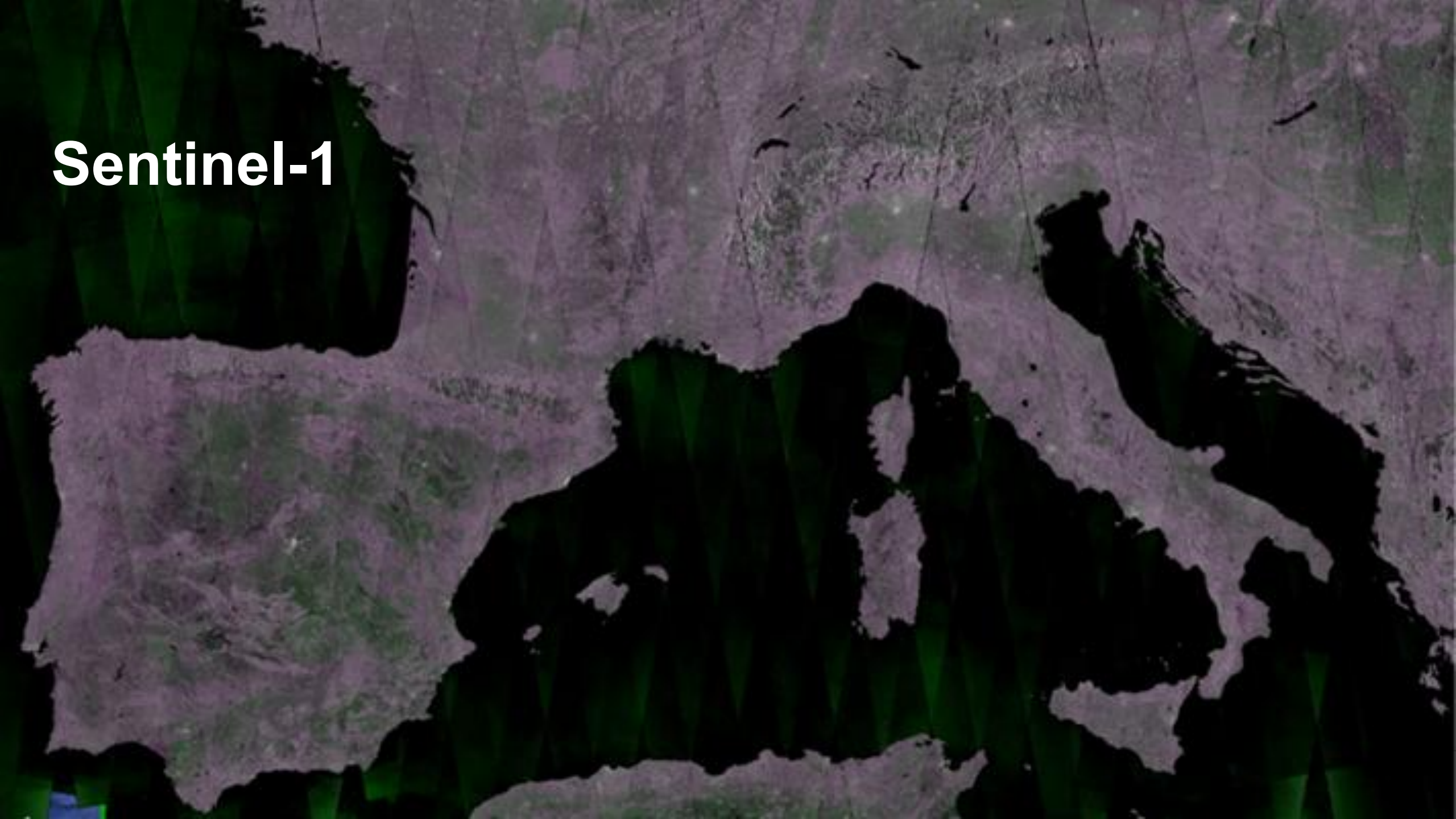
Terrain



Atmosphere

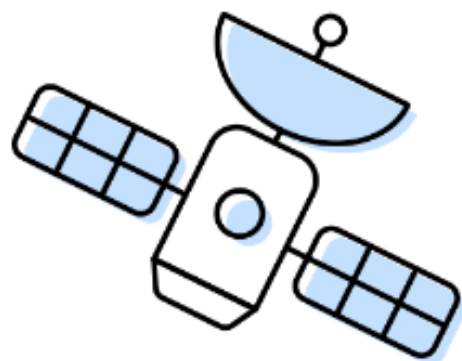


Sentinel-1



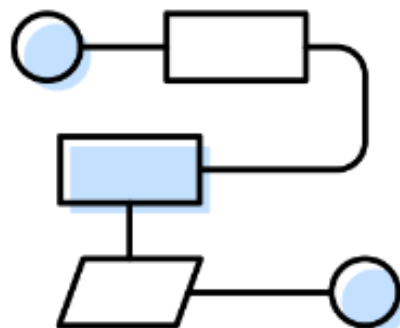
Meet Earth Engine

Google Earth Engine combines a multi-petabyte catalog of satellite imagery and geospatial datasets with planetary-scale analysis capabilities. Scientists, researchers, and developers use Earth Engine to detect changes, map trends, and quantify differences on the Earth's surface. Earth Engine is now available for commercial use, and remains free for academic and research use.



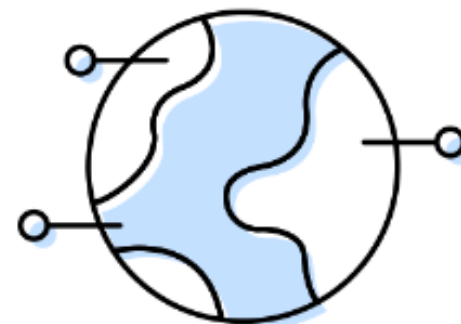
Satellite Imagery

+



Your Algorithms

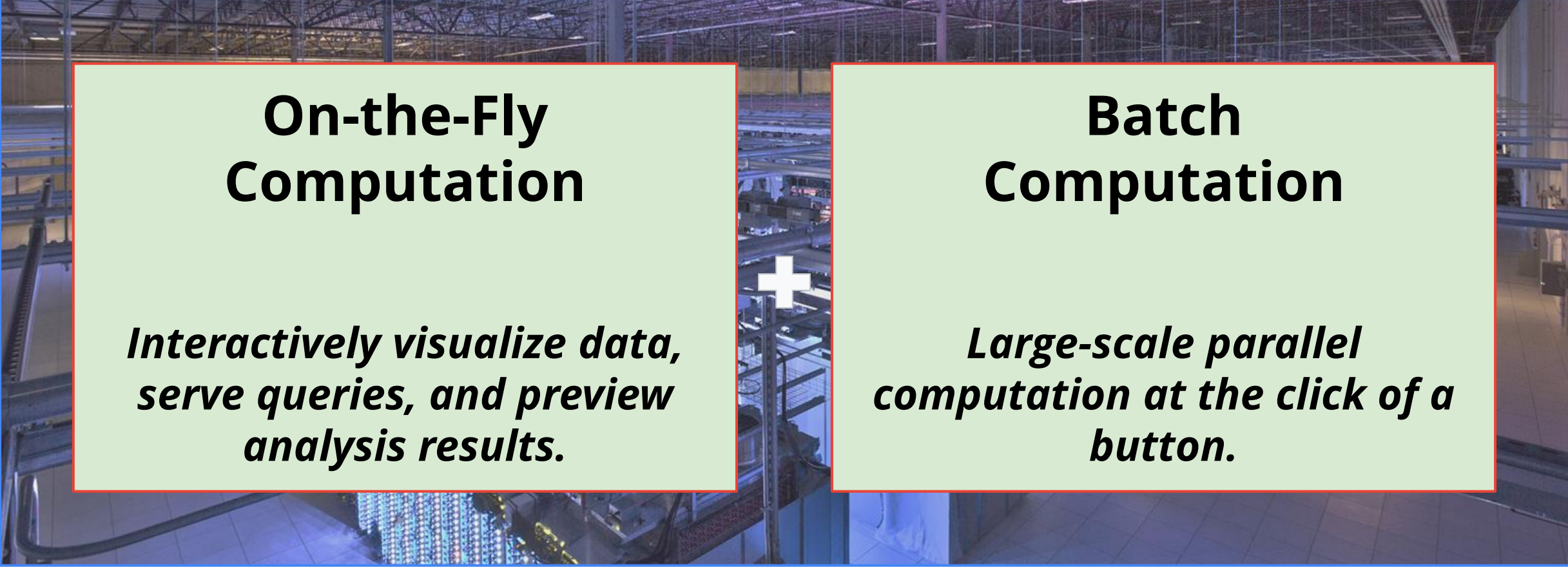
+



Real World Applications

Activate Wii
Go to Settings t

Computation Platform



On-the-Fly Computation

*Interactively visualize data,
serve queries, and preview
analysis results.*

+

Batch Computation

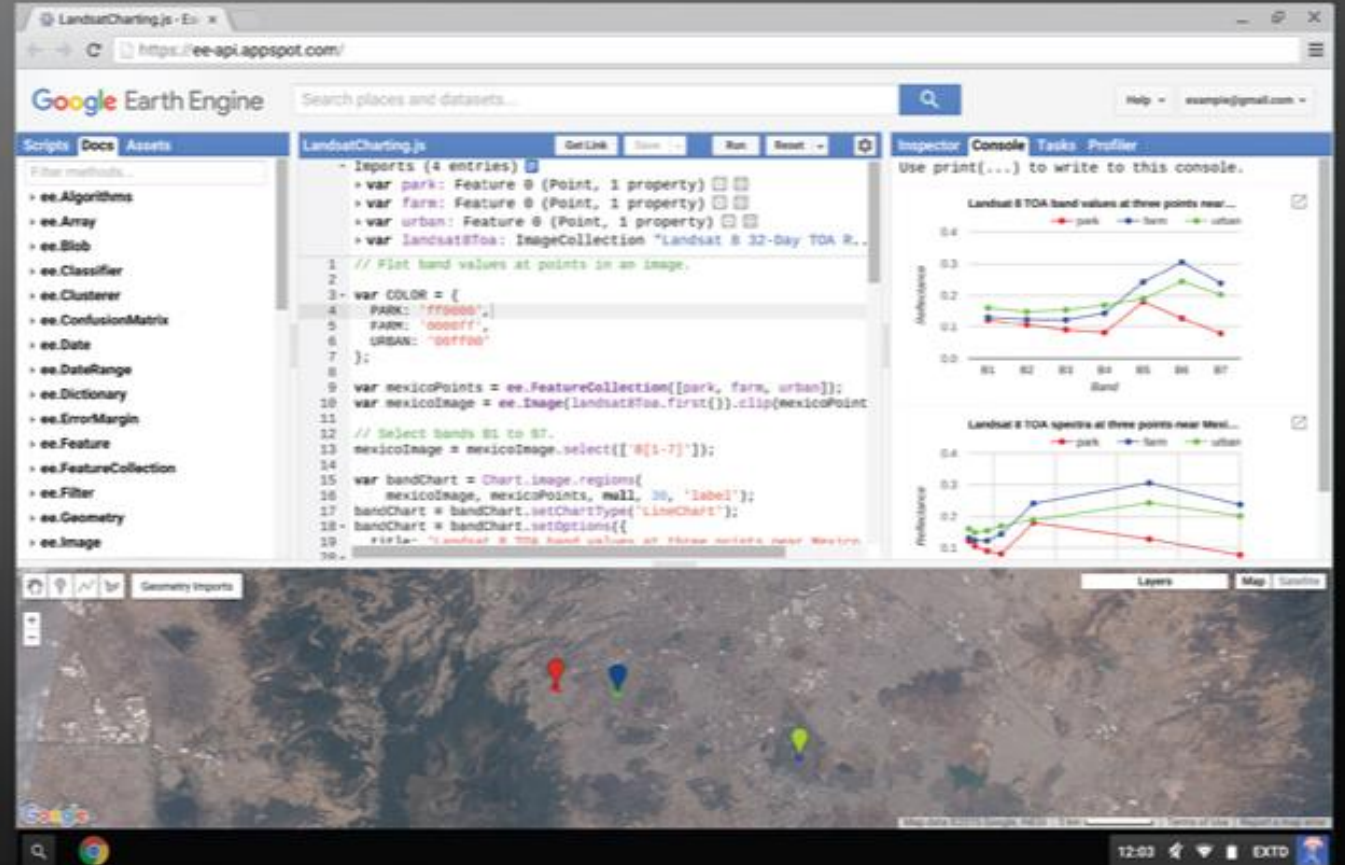
*Large-scale parallel
computation at the click of a
button.*

Powerful JavaScript* API

Convenient Tools

Use our web-based code editor for fast, interactive algorithm development with instant access to petabytes of data.

[Learn About The Code Editor](#)



Activate Windows
Go to Settings to activate Windows.

Javascript API

The Javascript API is a set of functions that allows you to interact with the data through a dedicated code editor.



1. The Earth Engine code editor



reflectance

Search!



Prod

Help

nclinton@google.com

PLACES

RASTERS

VNP09GA: VIIRS Surface **Reflectance** Daily 500m and 1km
 MOD0CGA.006 Terra Ocean **Reflectance** Daily Global 1km
 MYDOCGA.006 Aqua Ocean **Reflectance** Daily Global 1km
 USGS Landsat 8 Surface **Reflectance** Tier 1
 USGS Landsat 8 Surface **Reflectance** Tier 2
 USGS Landsat 4 Surface **Reflectance** Tier 1
 USGS Landsat 4 Surface **Reflectance** Tier 2
 USGS Landsat 5 Surface **Reflectance** Tier 1

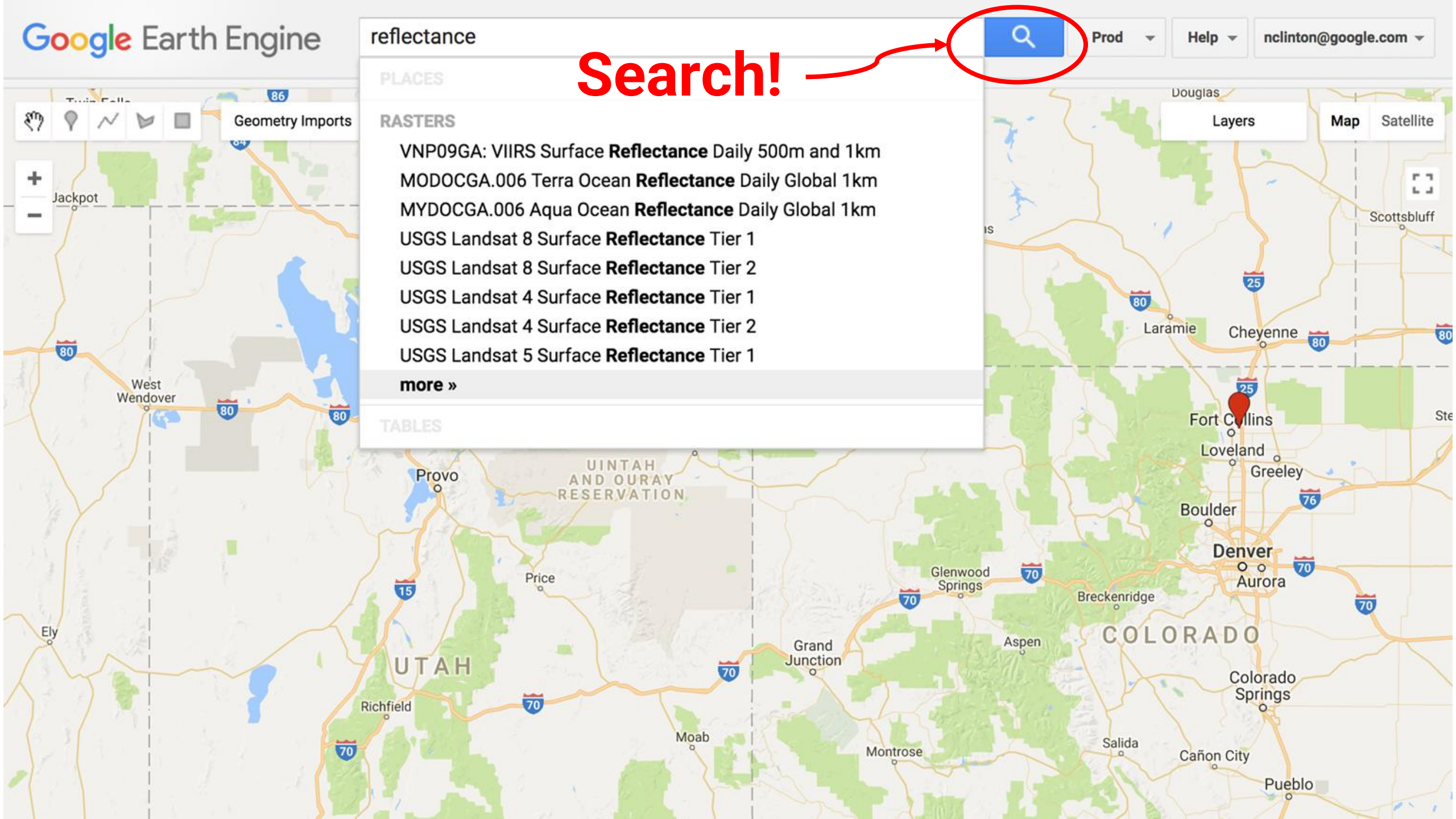
more »

TABLES

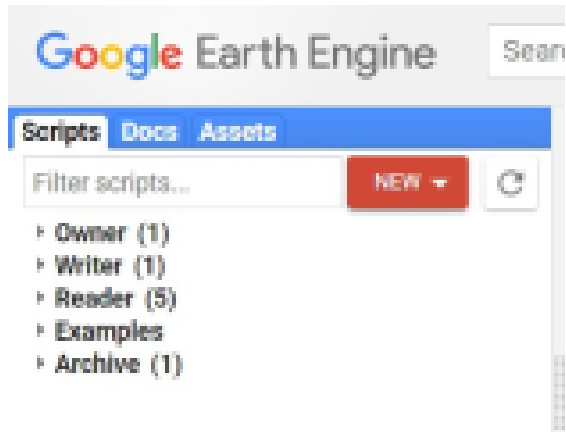
Layers

Map

Satellite



1

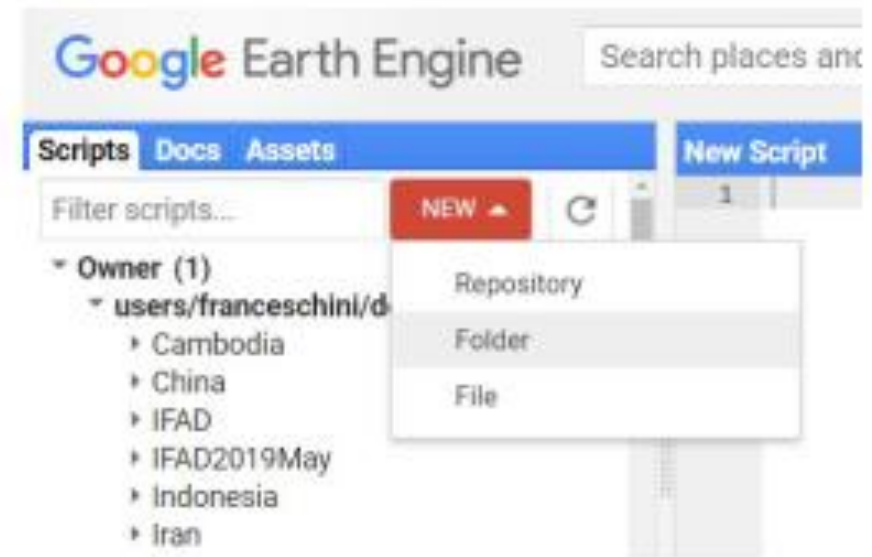


EXERCISE: CREATE A FOLDER

We will create a folder to store the scripts of the training:

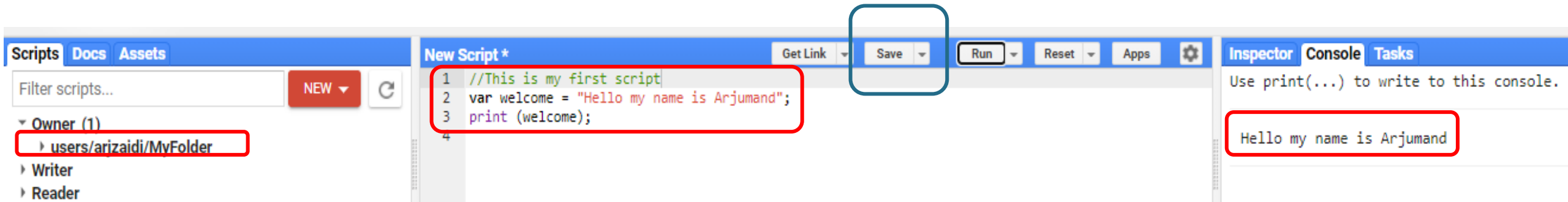
- Click on the NEW button
- Select Folder

2



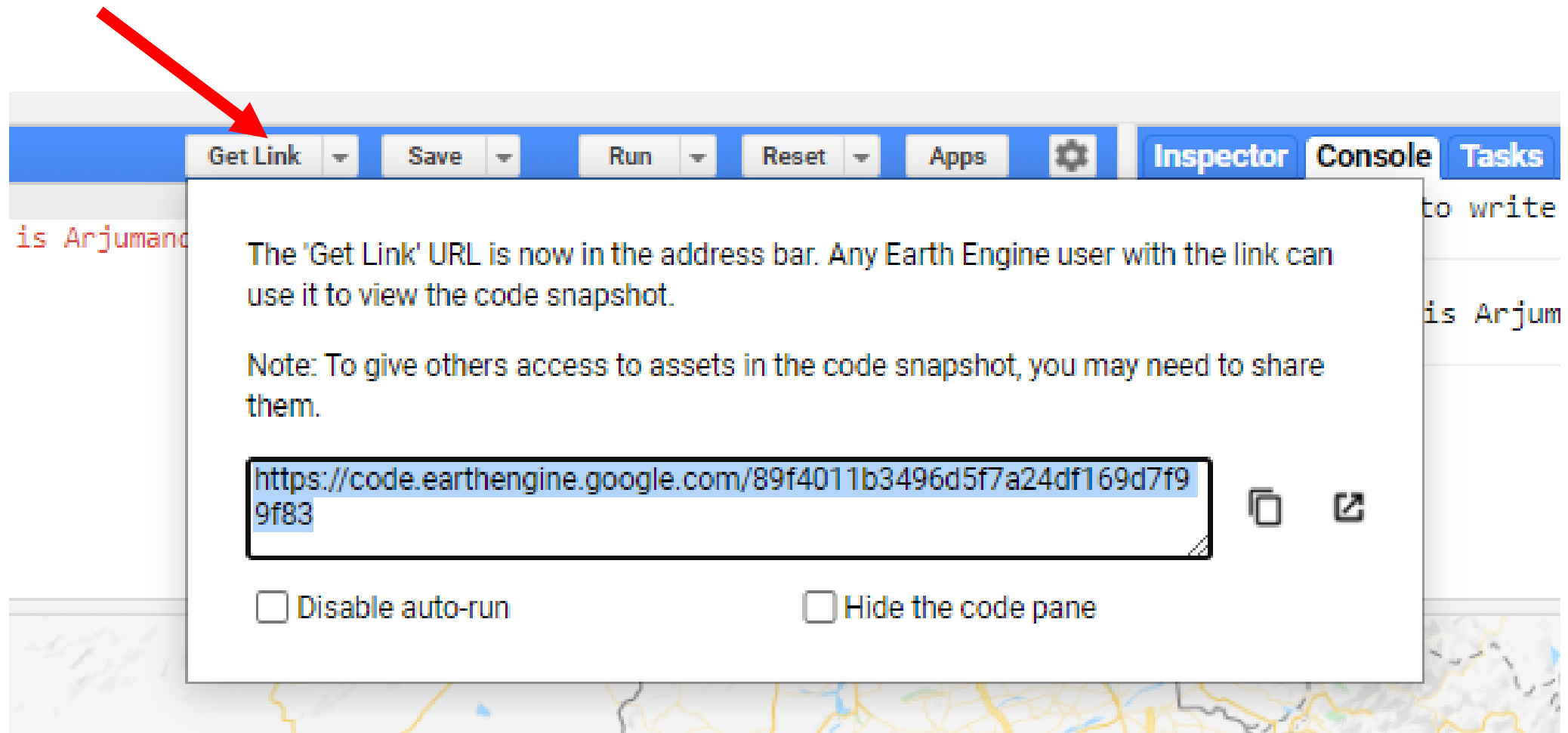
3 – Type name of the folder

Your first Script: Hello, I am 'your name'

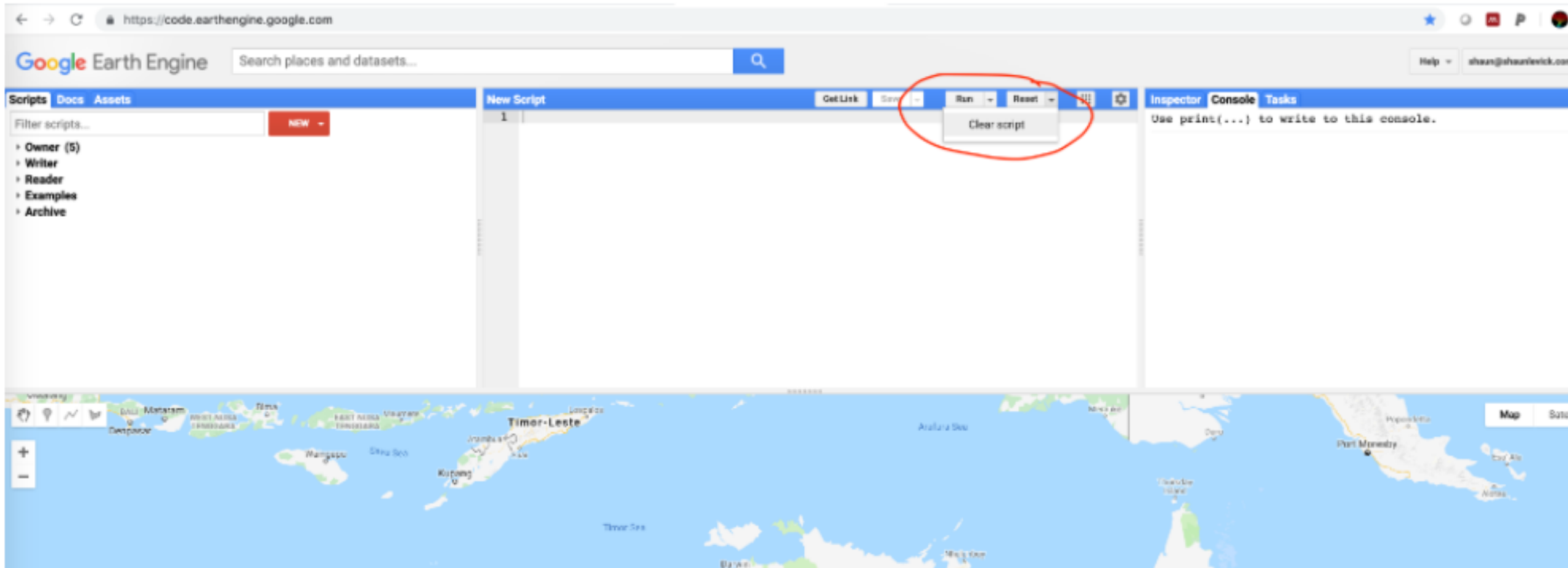


- Each variable has to be declared with the `var` operator
- Each line has to end with a semicolon
- Outputs are written in the Console panel
- Double slash is used to comment

The **Get Link** button, allows to generate an URL link of the script that can be shared with other users



Clear Script



Data Types and Geospatial Processing Functions

- **Image** - band math, clip, convolution, neighborhood, selection ...
- **Image Collection** - map, aggregate, filter, mosaic, sort ...
- **Feature** - buffer, centroid, intersection, union, transform ...
- **Feature Collection** - aggregate, filter, flatten, merge, sort ...
- **Filter** - by bounds, within distance, date, day-of-year, metadata ...
- **Reducer** - mean, linearRegression, percentile, histogram
- **Join** - simple, inner, outer, inverted ...
- **Kernel** - square, circle, gaussian, sobel, kirsch ...
- **Machine Learning** - CART, random forests, bayes, SVM, kmeans, cobweb ...
- **Projection** - transform, translate, scale ...

over 1000 data types and operators, and growing!

Global composites with a few lines of code

```
var composite = ee.Algorithms.Landsat.simpleComposite({  
  collection: ee.ImageCollection('LANDSAT/LC08/C01/T1'),  
  asFloat: true  
});
```

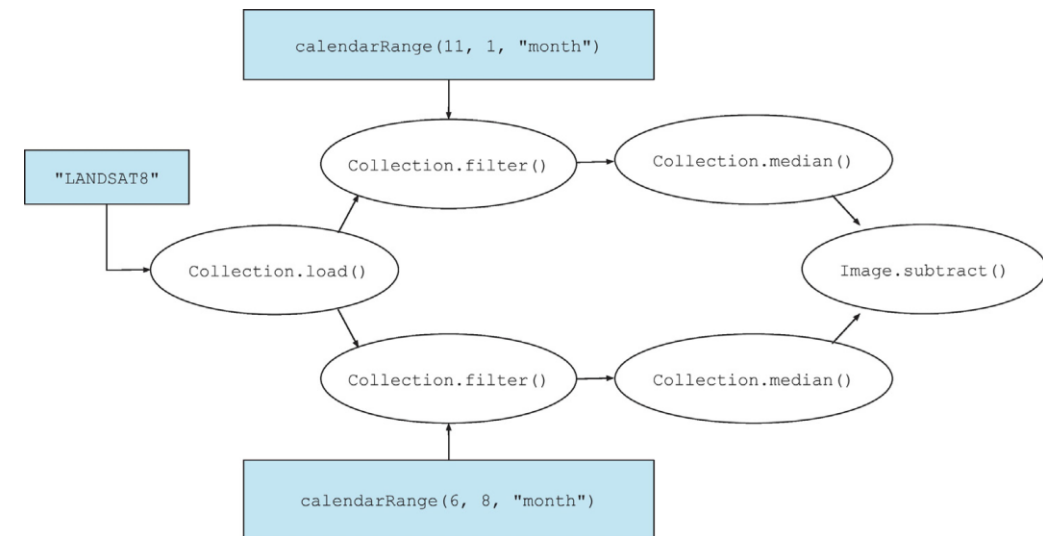
```
Map.addLayer(composite,  
  {bands: ['B4', 'B3', 'B2'], max: 0.3},  
  'composite');
```


Google Earth Engine: Planetary-scale geospatial analysis for everyone

Noel Gorelick^{a,*}, Matt Hancher^b, Mike Dixon^b, Simon Ilyushchenko^b, David Thau^b, Rebecca Moore^b

<https://www.sciencedirect.com/science/article/pii/S0034425717302900>

```
collection = ee.ImageCollection("LANDSAT8")
winter = collection.filter(ee.Filter.calendarRange(11, 1, "month"))
summer = collection.filter(ee.Filter.calendarRange(6, 8, "month"))
diff = summer.median().subtract(winter.median())
```



Data Catalog

The screenshot displays the Google Earth Engine web interface. At the top, a search bar contains the word "elevation", which is highlighted by a red rectangle. A red arrow points from the text "Or Search Keywords" below to this search bar. To the right of the search bar, a dropdown menu is open, also highlighted by a red rectangle. It contains two options: "View Data Catalog" and "Suggest a dataset". A red arrow points from the text "View" below to the "View Data Catalog" option. The interface includes a sidebar on the left with "Scripts", "Docs", and "Assets" tabs, and a main map area showing a geographical view of Iran, Pakistan, and parts of India. The bottom of the screen shows a Windows taskbar with various application icons and the system clock indicating 12:50 PM on 10/18/2020.

Google Earth Engine

elevation

Scripts Docs Assets

Filter scripts... NEW

Owner (1)
users/arjzaidi/MyFolder

Writer
Reader
Examples
Archive

New Script *

```
1 //This is my first script
2 var welcome = "Hello my name is Arjumand";
3 print(welcome);
4
```

Get Link Save

View Data Catalog
Suggest a dataset

Inspector Console Tasks

Use print(...) to write to this console.

Hello my name is Arjumand JSON

Or Search Keywords

View

Map Satellite

Iran Pakistan

Shiraz Kerman Zahedan Bandar Abbas Shiraz

Faisalabad Lahore Multan Bahawalpur Nawabshah

HIMACHAL PRADESH CHANDIGARH PUNJAB HARYANA UTTARAKHAND UTTAR PRADESH RAJASTHAN

New Delhi Agra Jaipur Lucknow

Go to Settings to activate Windows. Show all

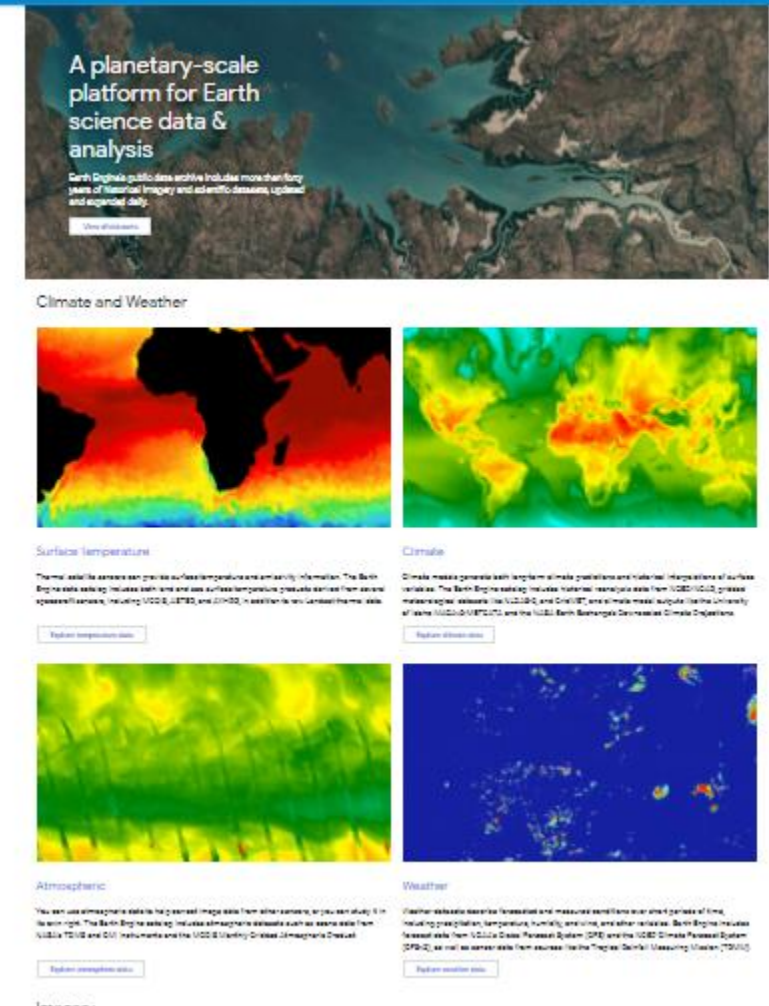
100666760-Forest....pdf Introductory cours....pdf

12:50 PM 10/18/2020

Data catalog contains a description of the available datasets by theme

<https://developers.google.com/earth-engine/datasets/>

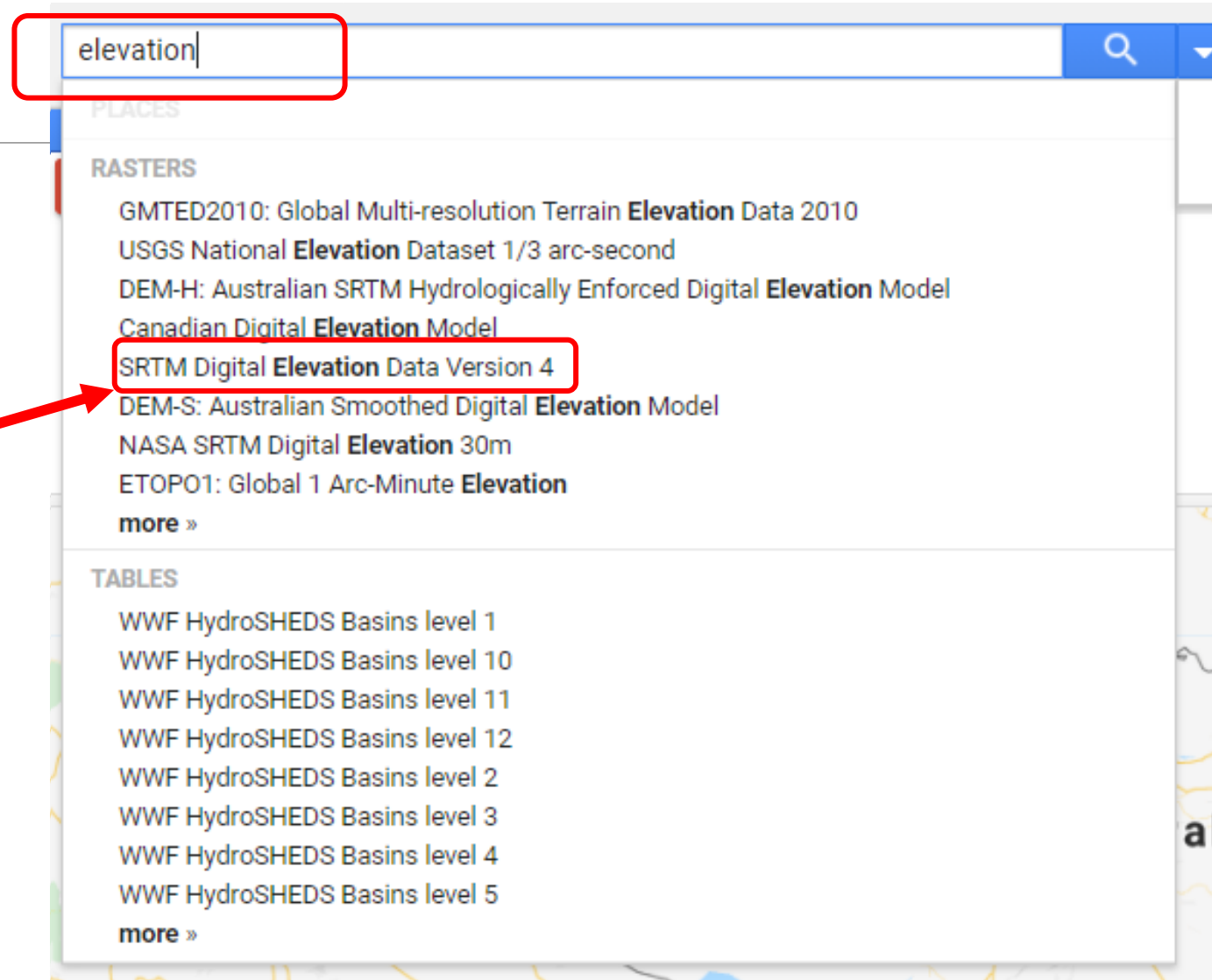
Search or directly select data from the catalog to import



Search Data - Elevation

Write key words in search window.

Select data



SRTM Digital Elevation Data Version 4



Dataset Availability

2000-02-11T00:00:00 - 2000-02-22T00:00:00

Dataset Provider

[NASA/CGIAR](#)

Collection Snippet

```
ee.Image("CGIAR/SRTM90_V4")
```

Tags

srtm elevation topography
dem geophysical cgiar

DESCRIPTION

BANDS

TERMS OF USE

CITATIONS

The Shuttle Radar Topography Mission (SRTM) digital elevation dataset was originally produced to provide consistent, high-quality elevation data at near global scope. This version of the SRTM digital elevation data has been processed to fill data voids, and to facilitate its ease of use.



Or- View Dataset in
Catalog Page

CLOSE

IMPORT

View Dataset in Catalog Page

SRTM Digital Elevation Data Version 4



Dataset Availability

2000-02-11T00:00:00 - 2000-02-22T00:00:00

Dataset Provider

[NASA/CGIAR](#)

Earth Engine Snippet

```
ee.Image("CGIAR/SRTM90_V4")
```

Tags

srtm

elevation

topography

dem

geophysical

cgiar

Description

[Bands](#)

[Terms of Use](#)

[Citations](#)

The Shuttle Radar Topography Mission (SRTM) digital elevation dataset was originally produced to provide consistent, high-quality elevation data at near global scale, and to facilitate its ease of use.

Explore in Earth Engine

```
var dataset = ee.Image('CGIAR/SRTM90_V4');  
var elevation = dataset.select('elevation');  
var slope = ee.Terrain.slope(elevation);  
Map.setCenter(-112.8598, 36.2841, 18);  
Map.addLayer(slope, {min: 0, max: 60}, 'slope');
```

[Open in Code Editor](#)

(long, lat, zoom level)

Open Code Editor

The screenshot displays the Google Earth Engine Open Code Editor interface. At the top, the Google Earth Engine logo is on the left, and a search bar with the placeholder text "Search places and datasets..." is in the center. To the right of the search bar are icons for help, chat, and user profile.

Below the search bar, there are tabs for "Scripts", "Docs", and "Assets". The "Scripts" tab is active, showing a list of scripts on the left. The script "CGIAR_SRTM90_V4" is selected, and its code is displayed in the center editor. The code is as follows:

```
1 var dataset = ee.Image('CGIAR/SRTM90_V4');
2 var elevation = dataset.select('elevation');
3 var slope = ee.Terrain.slope(elevation);
4 Map.setCenter(-112.8598, 36.2841, 10);
5 Map.addLayer(slope, {min: 0, max: 60}, 'slope');
6
```

At the top of the script editor, there are buttons for "Get Link", "Save", "Run", "Reset", "Apps", and a settings gear icon. The "Run" button is highlighted with a red rectangle.

On the right side of the interface, there are tabs for "Inspector", "Console", and "Tasks". The "Console" tab is active, showing the message "Use print(...) to write to this console."

Below the code editor, a map is displayed showing the visualization of the slope data. The map shows a topographic view of a region with a river network. A legend in the bottom right corner of the map shows a checkmark, the label "slope", a settings gear icon, and a color bar. This legend is also highlighted with a red rectangle.

SRTM Digital Elevation Data Version 4



DESCRIPTION

BANDS

TERMS OF USE

CITATIONS

The Shuttle Radar Topography Mission (SRTM) digital elevation dataset was originally produced to provide consistent, high-quality elevation data at near global scope. This version of the SRTM digital elevation data has been processed to fill data voids, and to facilitate its ease of use.

Dataset Availability

2000-02-11T00:00:00 - 2000-02-22T00:00:00

Dataset Provider

[NASA/CGIAR](#)

Collection Snippet

```
ee.Image("CGIAR/SRTM90_V4")
```

Tags

srtm

elevation

topography

dem

geophysical

cgiar

CLOSE

IMPORT

NEW ▾ ↻

New Script * Get Link ▾ Save ▾ Run ▾ Reset ▾ Apps ⚙

Imports (1 entry) ▾

- var image: Image "SRTM Digital Elevation Data Version 4" (1 band) ⚙

```
1 //DEM Data visualization
2
3
```

gine elevation 🔍 ▾

NEW ▾ ↻

New Script * Get Link ▾ Save ▾ Run ▾ Reset ▾ Apps ⚙

Imports (1 entry) ▾

- var DEM: Image "SRTM Digital Elevation Data Version 4" (1 band) ⚙

```
1 //DEM Data visualization
2
3
```

New Script * Get Link ▾ Save ▾ Run ▾ Reset ▾ Apps ⚙

Imports (1 entry) ▾

- var DEM: Image "SRTM Digital Elevation Data Version 4" (1 band) ⚙

```
1 //DEM Data visualization
2 print (DEM);
3
```

Inspector Console Tasks

Use print(...) to write to this console.

Image CGIAR/SRTM90_V4 (1 band) JSON

- type: Image
- id: CGIAR/SRTM90_V4
- version: 1595337443981112
- bands: List (1 element)
- properties: Object (24 properties)

Google Earth Engine

elevation

Scripts Docs Assets

Filter scripts... NEW

Owner (1)
+ users/arjzaidi/MyFolder
+ Writer
+ Reader
+ Examples
+ Archive

New Script *

Get Link Save Run Reset Apps

* Imports (1 entry)
+ var DEM: Image "SRTM Digital Elevation Data Version 4" (1 band)

```
1 //DEM Data visualization  
2 print(DEM);  
3 Map.addLayer(DEM, {min: 0, max: 3000}, 'Digital Elevation Model');  
4
```

Inspector Console Tasks

Use print(...) to write to this console.

+ Image CGIAR/SRTM90_V4 (1 band) 300M

Layers

✓ Digital Elevation Model

Map Satellite

Layer 1 visualization parameters

☒ 1 band (Grayscale) ☐ 3 bands (RGB)

elevation

Range

-41

-

5432

Stretch: 100%

Opacity

1.00

☒ Gamma

☐ Palette

1.00

Import

Apply

Close

Visualization parameters

☐ 1 band (Grayscale) ☐ 3 bands (RGB)

elevation

-

5432

Stretch: 100%

1.00

☐ Gamma

☒ Palette



Import

Apply

Close

Layer 1 visualization parameters

☒ 1 band (Grayscale) ☐ 3 bands (RGB)

elevation

Range

-41

-

5432

Stretch: 100%

Opacity

1.00

☐ Gamma

☒ Palette

Import

#ffffff

Ok

Cancel

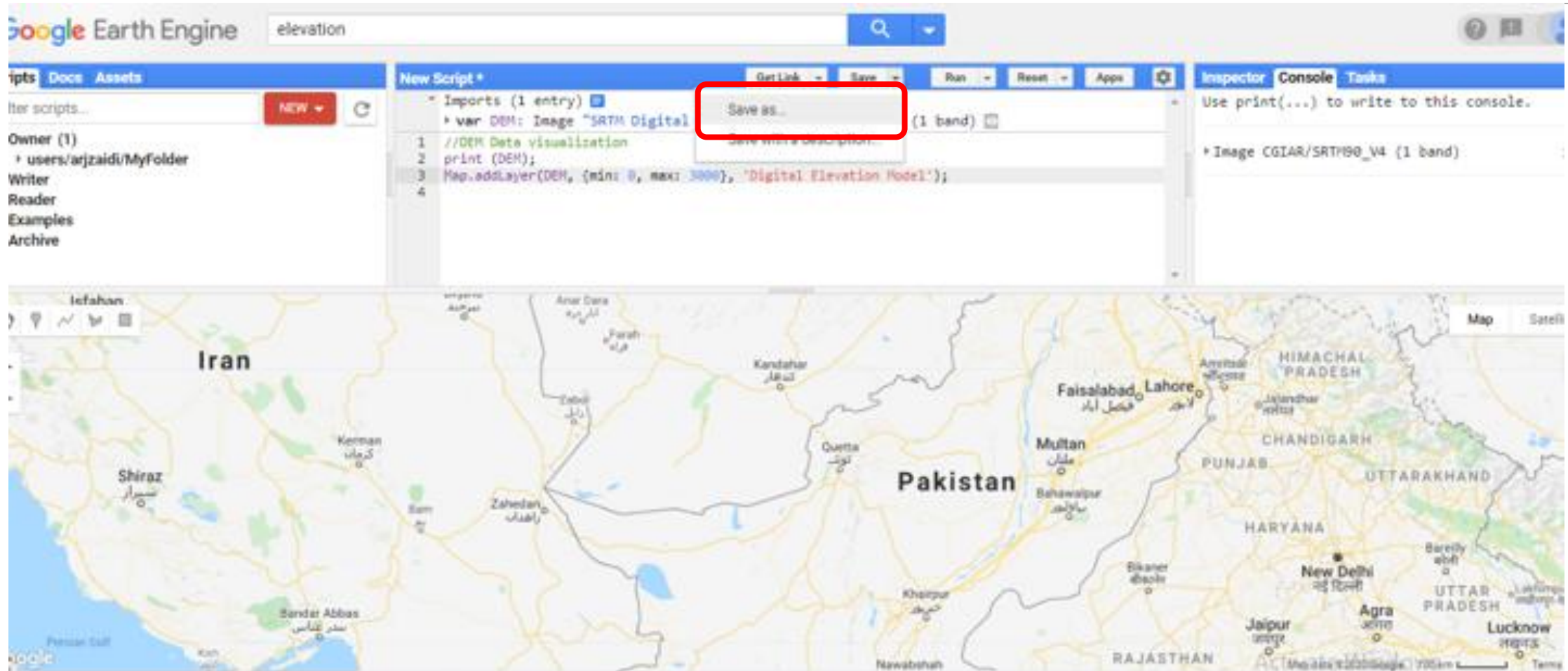
Change Symbology Color

The screenshot displays a web-based GIS application interface. The top section features a 'New Script *' editor with a toolbar containing 'Get Link', 'Save', 'Run', 'Reset', and 'Apps' buttons. The script editor shows the following code:

```
Imports (1 entry)
  var DEM: Image "SRTM Digital Elevation Data Version 4" (1 band)
1 //DEM Data visualization
2 print (DEM);
3 Map.addLayer(DEM, {min: 0, max: 3000}, 'Digital Elevation Model');
4 //Add DEM color
5 Map.addLayer(DEM, {min: 0, max: 3000, palette: ['Blue', 'Yellow', 'Red']}, 'Elevation above sea level');
6
```

The fifth line of code is highlighted with a red rectangular box. To the right of the script editor is the 'Inspector' and 'Console' panel. The 'Console' tab is active, displaying the text: 'Use print(...) to this console.' Below this, a tree view shows 'Image CGIAR/SRTM'. The bottom section of the interface shows a map visualization of a coastal area, with land areas colored in a gradient from blue (low elevation) to yellow and red (high elevation). A 'Layers' panel is visible in the bottom right corner of the map area.

Saving your Script



THANK YOU